

Long-Time Coherent Integration Enhancement of Moving Target Line Spectrum Based on Matched Frequency-Trajectory Estimation

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Correspondence: Zhenxing Zhao (zhaozx@ouc.edu.cn)**Received:** 16 June 2026 | **Revised:** 9 July 2026 | **Accepted:** 19 July 2026**Keywords:** acoustic signal detection | acoustic signal processing | frequency-trajectory estimation | long-time coherent integration

ABSTRACT

Line spectra radiated by moving targets are important features for passive sonar detection and classification. Relative motion between a target and the receiver induces time-varying Doppler shifts in the received line spectra, causing phase misalignment during long-time Fourier integration and resulting in spectral broadening, energy dispersion and coherent integration loss. To address this problem, this paper proposes a matched frequency-trajectory estimation method for long-time coherent integration enhancement (MFT-LTCIE). The method transforms the received signal into its analytic representation, extracts the narrow-band component around the target line spectrum and partitions the signal into short-time segments. For each segment, matching responses are computed over a candidate frequency grid, and a maximum-matching criterion is used to estimate the segment-wise frequency trajectory. The estimated trajectory is interpolated onto the sample-time axis and used to construct a phase-compensation function relative to a reference frequency. Trajectory-guided phase compensation restores phase coherence, and a full-duration FFT of the compensated signal yields an enhanced coherent integration spectrum. Simulations show that MFT-LTCIE suppresses Doppler-induced spectral broadening and maintains sharp spectral peaks under different signal-to-noise ratios and target speeds. Experimental ship-radiated noise data further demonstrate its ability to track line-spectrum variations and refocus dispersed energy.

1 | Introduction

Passive sonar detects, localises and classifies underwater targets by analysing acoustic signals. Narrowband line spectra are critical features of moving targets, such as ships and submarines, generated by mechanical vibrations, propeller rotation and propulsion systems. These spectra exhibit concentrated frequencies, long propagation distances and clear physical meaning, making them valuable for target detection [1, 2]. Basic analysis methods, including periodogram, power spectral estimation and LOFAR representations, reveal the energy distribution of target-radiated noise in frequency and time-frequency domains [3–5]. Under low signal-to-noise ratio (SNR) and complex ocean conditions,

weak line spectra are often obscured by background noise or interference, resulting in unclear spectral peaks, fragmented trajectories and increased false alarms [6, 7].

Extending observation time can improve frequency resolution for stationary targets. For moving targets, Doppler-induced frequency shifts cause spectral peak broadening and energy leakage during long-time coherent integration, reducing enhancement of weak line spectra. Existing approaches address this problem via long-time coherent integration, line-spectrum enhancement and trajectory tracking. Long-time coherent integration accumulates target energy over extended periods, improving weak-target

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detection [8, 9]. Compensation-based methods correct frequency drift or time variations to stabilise line spectra and improve spectral focussing [10–12]. Line-spectrum enhancement and extraction methods exploit sparsity or deep learning techniques to improve visibility and recovery of weak spectra [13–19]. Trajectory-tracking approaches, including track-before-detect (TBD), multi-frame coherent TBD, HMM and particle filtering methods, leverage spectral continuity for detection and tracking under low SNR [20–25]. Parameterised transforms, such as Radon–Fourier and PRPFT, provide motion compensation and energy accumulation but incur high computational cost when searching over velocity, acceleration or polynomial parameters [26, 27].

Despite these advances, two main challenges remain unresolved. First, most existing methods rely on either explicit motion parameter search or predefined motion models, which limits their applicability under complex or unknown target motion conditions. Second, trajectory-based or transform-based methods often suffer from high computational burden or sensitivity to parameter settings, especially in long-time observation scenarios with varying Doppler effects. To address these issues, this paper proposes a matched frequency-trajectory estimation method for long-time coherent integration enhancement (MFT-LTCIE). The key idea is to decompose the long-time observation into short-time segments, where the line spectrum is approximately stationary, and to estimate the frequency trajectory through a local matching criterion in a candidate frequency grid. Unlike conventional methods that rely on global parameter searches or explicit motion models, the proposed method directly constructs the frequency trajectory from data-driven segment-wise matching results, and further performs trajectory-guided phase compensation to achieve long-time coherent integration.

The main contributions of this work are summarised as follows:

1. A segment-wise matched frequency-trajectory estimation framework is developed to track Doppler-induced line-spectrum variations without requiring prior motion parameters.
2. A trajectory-guided phase compensation strategy is designed to restore phase coherence and enhance long-time coherent integration gain.
3. Both simulation and real ship-radiated noise experiments demonstrate that the proposed method achieves sharper spectral focussing and improved robustness under varying target speeds and SNR conditions.

The remainder of this paper is organised as follows: Section 2 introduces the signal model and problem formulation. Section 3 details the proposed MFT-LTCIE method. Section 4 presents the simulation results and comparative analysis. Section 5 provides experimental validation using ship-radiated noise data. Finally, Section 6 concludes the paper.

2 | Signal Model and Problem Formulation

2.1 | Line-Spectrum Signal Model

As shown in Figure 1, this paper considers a scenario in which a single hydrophone receives line spectra radiated by a moving

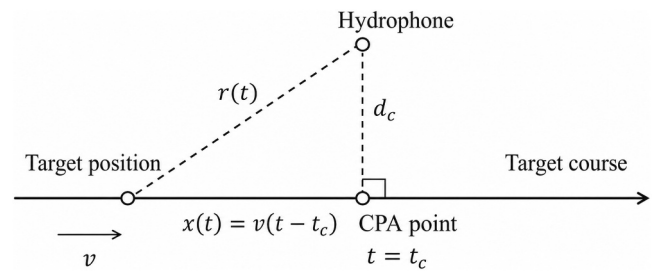


FIGURE 1 | Uniform linear motion model.

target. The hydrophone is assumed to be fixed near the coordinate origin, while the target moves with uniform linear motion and radiates a narrowband line spectrum. Let the reception time be denoted by t , and the corresponding emission time by τ . Due to the finite propagation speed of sound in water, the signal received at time t is not emitted by the target at the same instant, but rather at an earlier emission time τ .

Assume that the radiated line spectrum has a center frequency f_0 , amplitude A and initial phase ϕ_0 . The real-valued emitted single-tone signal can be expressed as follows:

$$s(\tau) = A \cos(2\pi f_0 \tau + \phi_0). \quad (1)$$

To facilitate phase compensation and long-time coherent integration, the real-valued signal is converted into its analytic representation as follows:

$$s_a(\tau) = A \exp(j(2\pi f_0 \tau + \phi_0)). \quad (2)$$

Let $r(\tau)$ denote the distance between the target and the hydrophone at the emission time $\tau(t)$, and let c denote the sound speed in water. The reception time and emission time satisfy the following relationship:

$$t = \tau + \frac{r(\tau)}{c}. \quad (3)$$

Therefore, the propagation delay is determined by the source–receiver distance at the emission time rather than by the distance at the reception time. Neglecting propagation loss and denoting the received amplitude by $A_r(t)$, the received analytic signal can be written as follows:

$$x_a(t) = A_r(t) \exp(j(2\pi f_0 \tau(t) + \phi_0)) + n_a(t), \quad (4)$$

where $n_a(t)$ represents additive noise in analytic form. According to Equation (4), the phase of the received signal is determined by the emission time $\tau(t)$. The received phase can be expressed as follows:

$$\phi(t) = 2\pi f_0 \tau(t) + \phi_0. \quad (5)$$

The instantaneous frequency is equal to the time derivative of the phase divided by 2π , yielding the following:

$$f_r(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt} = f_0 \frac{d\tau(t)}{dt}. \quad (6)$$

Equation (6) shows that the variation of the received frequency for a moving target fundamentally originates from the nonlinear relationship between the emission time $\tau(t)$ and the reception

time t . In the following sections, the quantity to be estimated and compensated is the received frequency trajectory $f_r(t)$, which is jointly determined by target motion and acoustic propagation.

2.2 | Doppler Variation Under Uniform Linear Motion

To describe the Doppler frequency variation caused by a moving target, this paper adopts a uniform linear motion model. The target moves along a straight line with velocity v , while the hydrophone remains fixed. At the emission time t_e , the target passes the position closest to the hydrophone, denoted by a distance d_c . Let the target motion be along the x -axis; the distance from the target to the hydrophone is then

$$r(\tau) = \sqrt{d_c^2 + v^2(\tau - t_c)^2}. \quad (7)$$

Substituting Equation (7) into the propagation relation yields the following:

$$t = \tau + \frac{1}{c} \sqrt{d_c^2 + v^2(\tau - t_c)^2}. \quad (8)$$

Solving Equation (8) as a causal relationship provides the emission time $\tau(t)$ corresponding to a given reception time t . In practice, the emission time can be determined from the reception time to reconstruct the received moving-target line spectrum as follows:

$$\tau(t) = \frac{c^2 t - v^2 t_c - \sqrt{c^2 v^2 (t - t_c)^2 + c^2 d_c^2 - v^2 d_c^2}}{c^2 - v^2}. \quad (9)$$

From the geometric relationship, the radial velocity can be expressed as the derivative of distance with respect to the emission time as follows:

$$v_r(\tau) = \frac{dr(\tau)}{d\tau} = \frac{v^2(\tau - t_c)}{r(\tau)}. \quad (10)$$

Therefore, the received frequency can be approximated as follows:

$$f_r(t) = \frac{f_0}{1 + v_r(\tau(t))/c}. \quad (11)$$

As shown in Figure 2, when the target approaches the hydrophone, the distance decreases and the received frequency is generally higher than f_0 . Near the closest point of approach, the radial velocity approaches zero, and the received frequency approaches f_0 . When the target moves away from the hydrophone, the distance increases and the received frequency is generally lower than f_0 . Hence, even if the target emits a single-tone signal, the received signal experiences a time-varying frequency trajectory due to the Doppler effect.

For stationary targets or frequency-stable targets, applying FFT or periodogram to the full-length data allows the signal energy to concentrate near a single frequency. When the target moves, the instantaneous received frequency $f_r(t)$ varies over time. Direct application of FFT on full-length data effectively attempts

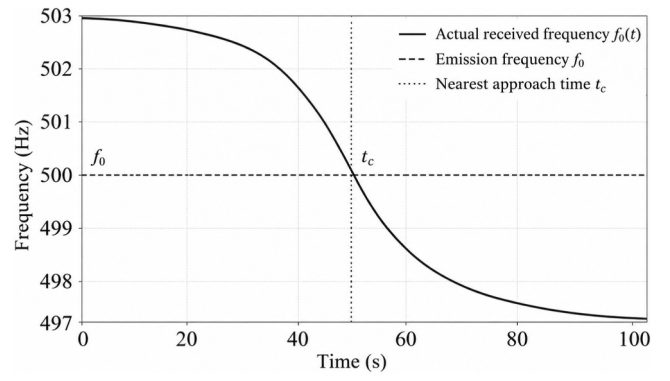


FIGURE 2 | True received frequency trajectory under uniform linear motion.

to match a fixed frequency to a varying frequency trajectory, leading to spectral peak broadening and energy leakage. For weak line signals, the main peak amplitude decreases and may be submerged in noise, making stable detection difficult via periodogram or LOFAR representations.

3 | Proposed MFT-LTCIE Method

3.1 | Overall Procedure

For narrowband line spectra of moving targets observed by a single hydrophone, relative motion between the target and the receiver causes the received frequency to vary with time, forming a frequency trajectory rather than a fixed spectral line. Direct full-duration Fourier integration spreads the spectral energy along the frequency axis and causes energy leakage, which reduces the spectral-peak amplitude and limits the long-time coherent integration gain. To address this problem, this paper proposes a matched frequency-trajectory estimation method for long-time coherent integration enhancement (MFT-LTCIE).

The proposed method proceeds as follows: first, the received signal is transformed into its analytic representation, and the narrowband component around the expected target frequency is extracted to suppress irrelevant components and broadband noise that may affect trajectory estimation. Next, a candidate frequency grid is constructed around the target line spectrum, and matching scores are computed over the grid for each short-time segment. A segment-wise maximum-matching criterion is then applied to obtain the frequency trajectory, which is interpolated onto the sample-time axis to form the full-duration trajectory. Finally, a phase-compensation function is constructed according to the frequency deviation relative to a reference frequency, and trajectory-guided phase correction is performed over the full observation interval. After compensation, the Doppler-dispersed line-spectrum energy is refocused around the reference frequency, and a full-duration FFT of the compensated signal yields an enhanced long-time coherent integration spectrum. The overall procedure is summarised in Table 1.

To further clarify the computational efficiency of the proposed method, the computational cost of MFT-LTCIE consists of three main parts: segment-wise local matching, trajectory estimation,

and FFT-based spectrum computation. The detailed comparison of computational complexity is summarised in Table 2.

Since the proposed method does not require exhaustive searches over motion-related parameters, such as velocity or acceleration grids, its computational burden is mainly determined by frequency-domain matching within a predefined candidate band. Therefore, compared with PRPFT and LTCILSD, which rely on large-scale parameter searching, the proposed method achieves higher computational efficiency while maintaining effective long-time coherent integration performance.

3.2 | Short-Time Spectrum Construction

According to the processing procedure of MFT-LTCIE, the first step is to select a candidate frequency band and construct segmental spectra. Assume that the target line spectrum is located within the candidate frequency band $[f_{\min}, f_{\max}]$. The reference centre frequency is defined as follows:

$$f_{\text{ref}} = \frac{f_{\min} + f_{\max}}{2}. \quad (12)$$

Let the original discrete received signal be denoted by $x[n]$. The input real-valued signal is first converted into its analytic form, which can be expressed as follows:

$$x_a[n] = x[n] + j\mathcal{H}\{x[n]\}, n = 0, 1, \dots, N-1, \quad (13)$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform. Since the target signal is a narrowband line spectrum, the narrowband complex signal corresponding to the selected frequency band is extracted

and denoted by $x_b[n]$. Subsequently, $x_b[n]$ is divided into overlapping segments using an analysis window of length L and a hop size D . The k -th segment is given by

$$s_k[n] = x_b[n + (k-1)D]w[n], n = 0, 1, \dots, L-1, \quad (14)$$

where $w[n]$ denotes a Hann window. Within each analysis window, the instantaneous received frequency of the moving target can be approximated as a constant. Therefore, the segmental processing transforms the estimation of a time-varying frequency trajectory into a series of local single-frequency estimation problems.

The selection of the analysis window length is guided by the quasi-stationary assumption of the line spectrum within each segment. Specifically, the window length L is chosen such that the maximum frequency variation within a segment remains below the preset frequency resolution threshold, ensuring that the target line spectrum can be approximated as a single-frequency component within each window. The hop size D is selected to guarantee sufficient continuity of the estimated frequency trajectory, while avoiding excessive computational cost. The frequency step is determined according to the expected Doppler variation range and the required resolution of trajectory estimation.

3.3 | Frequency Trajectory Estimation

To estimate the frequency trajectory of the target line spectrum, a candidate frequency set F is constructed within the candidate frequency band $[f_{\min}, f_{\max}]$ using a frequency step Δf as follows:

TABLE 1 | Procedure of the MFT-LTCIE method.

Input: Received signal $x[n]$, candidate frequency band $[f_{\min}, f_{\max}]$, segment length L , hop size D , frequency step Δf	
Output: Estimated frequency trajectory $\hat{f}[n]$, compensated signal $z[n]$, the MFT-LTCIE spectrum	
Start	
1. Convert $x[n]$ into its analytic form and extract the narrowband complex signal $y[n]$ within the candidate frequency band.	
2. Segment $y[n]$, compute local matching scores for all segments and candidate frequencies, and construct the normalised score map $\hat{S}_k(f_m)$.	
3. Estimate the optimal segment-wise frequency trajectory $\{\hat{f}_k\}$ according to the maximum-matching criterion.	
4. Interpolate $\{\hat{f}_k\}$ to obtain the sample-level frequency trajectory $\hat{f}[n]$.	
5. Construct the compensation phase and perform phase compensation on $x_b[n]$ to obtain $z[n]$.	
6. Compute the long-time spectrum of $z[n]$ and obtain the 1-D spectrum:	
$P_{\text{MFT-LTCIE}}(f) = \left \sum_{n=0}^{N-1} z[n]w_s[n]\exp(-j2\pi\frac{fn}{f_s}) \right ^2, f \in [f_{\min}, f_{\max}]$	
end	

TABLE 2 | Computational complexity comparison of different methods.

Method	Multiplications	Additions
Periodogram	$0.5N \log_2 N$	$N \log_2 N$
PRPFT	$0.5N \log_2 M + N_{\alpha_1}N_{\alpha_2}K(0.5 \log_2 K + 1)$	$N \log_2 M + N_{\alpha_1}N_{\alpha_2}K \log_2 K$
LTCILSD	$N \log_2 M + N_{\beta_1}N_{\alpha_2}N + 0.5N_{\alpha_1}N_{\alpha_2} \log_2 N_{\alpha_1}$	$2N \log_2 M + N_{\alpha_1}N_{\alpha_2} \log_2 N_{\alpha_1}$
MFT-LTCLE	$KMN_s + N + 0.5N \log_2 N$	$KMN_s + KM + N + N \log_2 N$

$$F = \{f_m = f_{\min} + (m - 1)\Delta f, m = 1, 2, \dots, M\}. \quad (15)$$

For the k -th segment and candidate frequency f_m , the matching between the candidate frequency and the segmental signal is quantified using a continuous-wave matched filter as follows:

$$S(k, m) = \left| \sum_{n=0}^{L-1} s_k[n] \exp\left(-j2\pi \frac{f_m n}{f_s}\right) \right|. \quad (16)$$

Due to variations in the distance between the target and the receiver, propagation loss and possible local noise fluctuations, the total energy of different short-time segments may differ. To reduce the impact of amplitude differences on trajectory estimation, the matching score of each frame is normalised as follows:

$$\tilde{S}(k, m) = \frac{S(k, m)}{\max_m [S(k, m)]} \quad (17)$$

The maximum value is then selected greedily. The frequency corresponding to the normalised maximum score is chosen as the segment-wise frequency as follows:

$$m_k^* = \arg \max_m \tilde{S}(k, m), f_k = f_{m_k^*}. \quad (18)$$

Considering that the target frequency usually varies smoothly over time, short-term outliers caused by local random peaks can be mitigated using a moving-median smoothing operation, which improves the continuity of the segmental trajectory without altering its global trend. By interpolating the segmental frequency trajectory at the centre time of each segment to the sampling instants, the full-duration frequency trajectory is obtained as follows:

$$\hat{f}[n] = \text{interp1}(t_k, f_k; t_n). \quad (19)$$

3.4 | Phase Compensation and Coherent Integration

After estimating the frequency trajectory, the next step is to construct the phase-compensation function based on the trajectory. Let f_{ref} be the reference frequency and $\hat{f}[n]$ denote the full-duration estimated frequency trajectory. The frequency deviation is then given as follows:

$$\Delta f[n] = \hat{f}[n] - f_{\text{ref}}. \quad (20)$$

Since frequency is the derivative of phase with respect to time, the time integral of the frequency deviation corresponds to the extra phase of the target signal relative to the reference frequency. Under discrete-time conditions, the accumulated phase corresponding to the frequency deviation is expressed as follows:

$$\phi_c[n] = 2\pi \frac{1}{f_s} \sum_{i=0}^n \Delta f[i]. \quad (21)$$

Let $x_b[n]$ denote the narrowband signal for the final compensation. The compensated signal $z[n]$ can be written as follows:

$$z[n] = x_b[n] \exp(-j\phi_c[n]), \quad (22)$$

where the exponential term provides reverse phase correction. If the frequency trajectory estimation is accurate, the compensated spectrum is effectively aligned with the reference frequency, and the phase of different time segments is realigned, providing the necessary conditions for long-time coherent integration.

Once phase compensation is completed, the compensated signal $z[n]$ is analysed over the full observation duration to obtain the long-time coherent integration spectrum as follows:

$$P_{\text{MFT-LTCIE}}(f) = \left| \sum_{n=0}^{N-1} z[n] w_s[n] \exp\left(-j2\pi \frac{fn}{f_s}\right) \right|^2, \quad (23)$$

$$f \in [f_{\min}, f_{\max}],$$

where $w_s[n]$ is a window function. This spectrum represents the output of MFT-LTCIE. In summary, the MFT-LTCIE method converts the time-varying Doppler frequency of the moving target into a series of short-time single-frequency estimation problems. By greedily selecting the maximum matching score for each segment and combining phase compensation, long-time coherent integration is restored. The method avoids global searches over velocity and acceleration, and its implementation is relatively simple.

4 | Simulation Results

4.1 | Performance Under Different SNRs

To evaluate the ability of the proposed MFT-LTCIE method to enhance line spectra of moving targets under different signal-to-noise ratio (SNR) conditions, the target speed and observation duration were set to 5 m/s and 300 s, respectively. Simulations were conducted at two SNR levels: 10 and 0 dB. The 10 dB case represents a relatively high-SNR condition, in which the target line spectrum is clearly visible and the frequency trajectory is relatively stable. The 0 dB case represents a low-SNR condition, in which the target line spectrum is strongly masked by background noise but still retains partial time-frequency continuity in the LOFAR representation, allowing trajectory estimation over a candidate frequency grid. In this simulation, the sampling frequency was set to 1000 Hz, the target centre frequency was 100 Hz, the sound speed was 1500 m/s and the closest point of approach (CPA) range was 1000 m. Figure 3 shows the target motion geometry. Figures 4 and 5 show the LOFAR representation, the estimated frequency trajectory and the enhanced spectra obtained using different methods.

For the proposed method, the analysis window length is set to 10 s with a hop size of 1 s, corresponding to a 90% overlap. The candidate frequency resolution is set to 0.001 Hz in both simulation and experimental studies. These settings are chosen to ensure sufficient time-frequency resolution for tracking the Doppler-induced line-spectrum variation while maintaining computational efficiency. For the comparison methods, PRPFT and LTCILSD are implemented with their recommended parameter settings in the literature [10]. Specifically, for PRPFT, the search step sizes are set to $\alpha_{1_step} = 0.1$ and $\alpha_{2_step} = 0.0005$. For LTCILSD, the parameters are set to $\alpha_{1_step} = 0.1$, $\alpha_{2_step} = 0.0005$ and $\beta_{1_step} = 0.001$. These settings ensure a

consistent resolution level in the parameter search space for fair performance comparison.

As shown in Figure 3, the target moves along a straight path at a constant speed and passes near the receiver during the

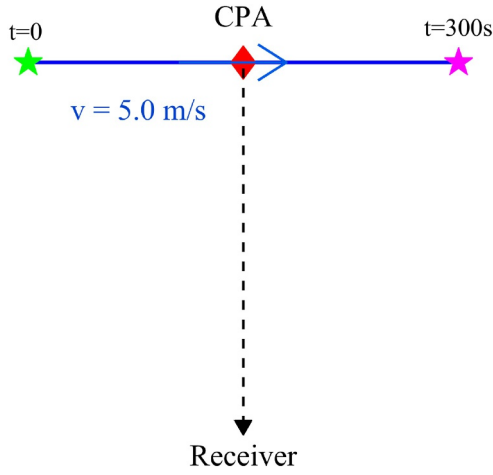
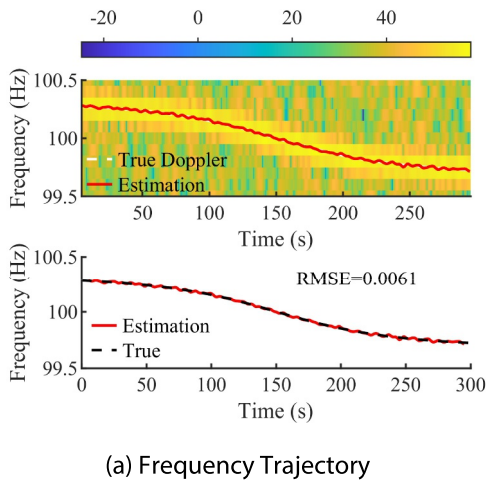


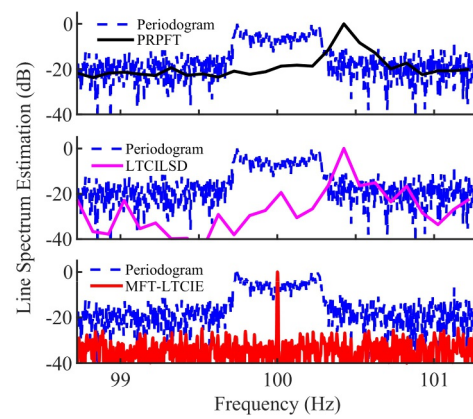
FIGURE 3 | Target motion relative to the receiver.

observation interval. The CPA is the point on the target track closest to the receiver, and the CPA range corresponds to the minimum slant range between the target and the receiver. Because the radial velocity of the target relative to the receiver varies continuously during the observation interval, the received line-spectrum frequency exhibits time-varying Doppler shifts. As the target approaches the CPA, the change in radial velocity becomes more pronounced, and the corresponding frequency trajectory becomes nonstationary. This motion geometry explains the time-varying line-spectrum trajectory observed in the LOFAR representation and the spectral energy spreading observed in the frequency-domain spectrum.

Under the 10 dB SNR condition, the LOFAR representation in Figure 4a clearly shows the time-varying frequency trajectory of the target line spectrum. Because of Doppler-induced frequency variation, the target energy is not concentrated at a single frequency but spreads over a finite frequency range. As shown in Figure 4a, MFT-LTCIE accurately estimates the frequency trajectory and follows the true trajectory, with the estimated trajectory agreeing well with the true frequency trajectory. This result indicates that the proposed method can capture the time-varying frequency behaviour of the moving target.

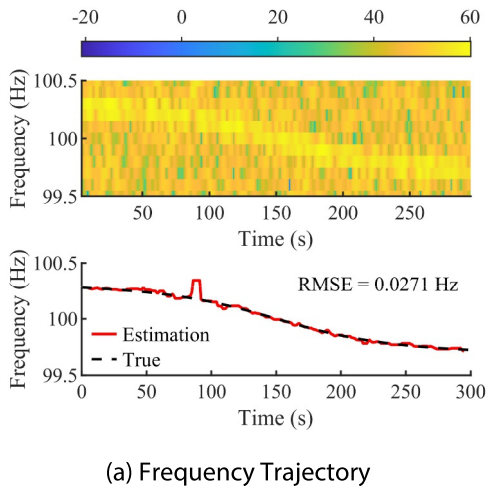


(a) Frequency Trajectory

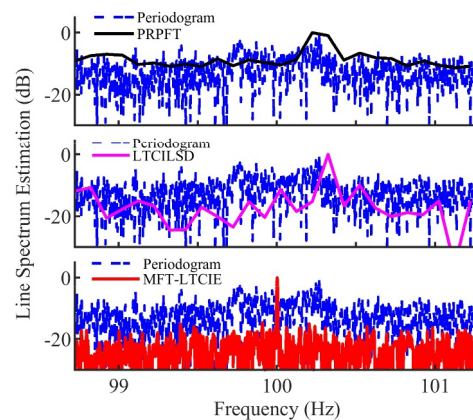


(b) One-Dimensional Spectrum

FIGURE 4 | MFT-LTCIE processing results at 10 dB SNR.



(a) Frequency Trajectory



(b) One-Dimensional Spectrum

FIGURE 5 | MFT-LTCIE processing results at 0 dB SNR.

The enhanced spectra in Figure 4b further illustrate the effectiveness of the proposed method. Although the conventional periodogram shows partial energy concentration around the target frequency, its spectral peak remains broad and poorly focused. In contrast, after frequency-trajectory estimation and phase compensation, MFT-LTCIE produces a sharper and more distinct spectral peak around the target centre frequency. This indicates that the proposed method can refocus the Doppler-dispersed line-spectrum energy and improve spectral focussing and coherent integration gain.

As shown in Figure 5, under the 0 dB SNR condition, background noise becomes more pronounced in the LOFAR representation, reducing the local continuity and clarity of the target line-spectrum trajectory. Nevertheless, the trajectory still retains partial time–frequency continuity, allowing MFT-LTCIE to estimate the trajectory over the candidate frequency grid. Compared with the 10 dB case, the estimated trajectory at 0 dB exhibits larger local fluctuations, and the enhanced spectral peak becomes slightly less prominent. Nevertheless, the main peak remains at a stable frequency and shows better spectral concentration than the original periodogram. These results indicate that the proposed method remains applicable under low-SNR conditions.

The results under both SNR conditions indicate that Doppler-induced frequency variation due to target motion disrupts the phase coherence required for direct long-time Fourier integration, leading to spectral-peak broadening, energy dispersion and stronger background fluctuations in the periodogram. By estimating the target frequency trajectory on a segment-wise basis and applying phase compensation guided by the estimated trajectory, MFT-LTCIE refocuses the Doppler-dispersed line-spectrum energy, thereby improving spectral concentration and peak resolvability.

4.2 | Performance Under Different Target Velocities

To further evaluate the robustness of the proposed MFT-LTCIE method under different Doppler-variation levels, simulations were conducted at an SNR of 10 dB with an observation duration of 300 s. The target speed was set to 0, 5, 10 and 15 m/s. The line-spectrum enhancement performance of the conventional periodogram, PRPFT, LTCILSD and MFT-LTCIE was compared. Figures 6–9 show the simulation results for different target speeds, including the LOFAR representations, estimated frequency trajectories and one-dimensional enhanced spectra

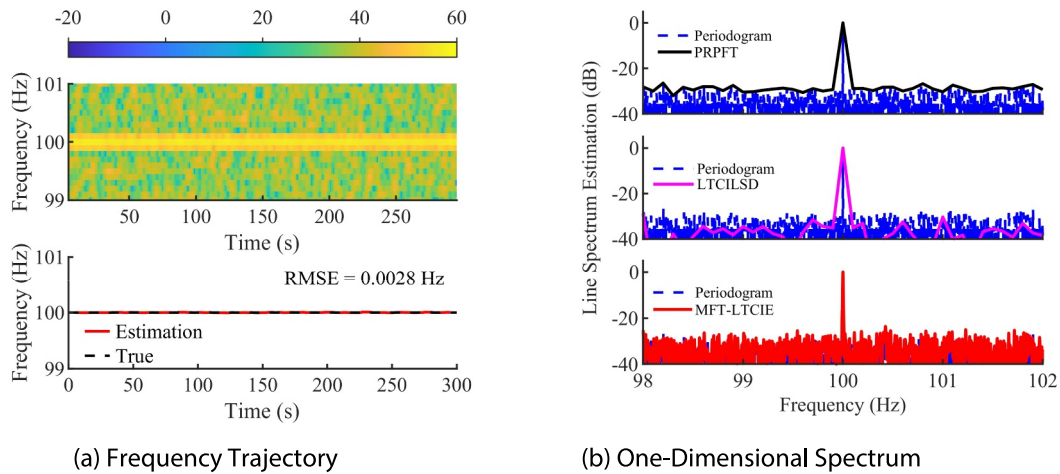


FIGURE 6 | Comparison of line-spectrum enhancement results obtained by different methods at a target velocity of 0 m/s.

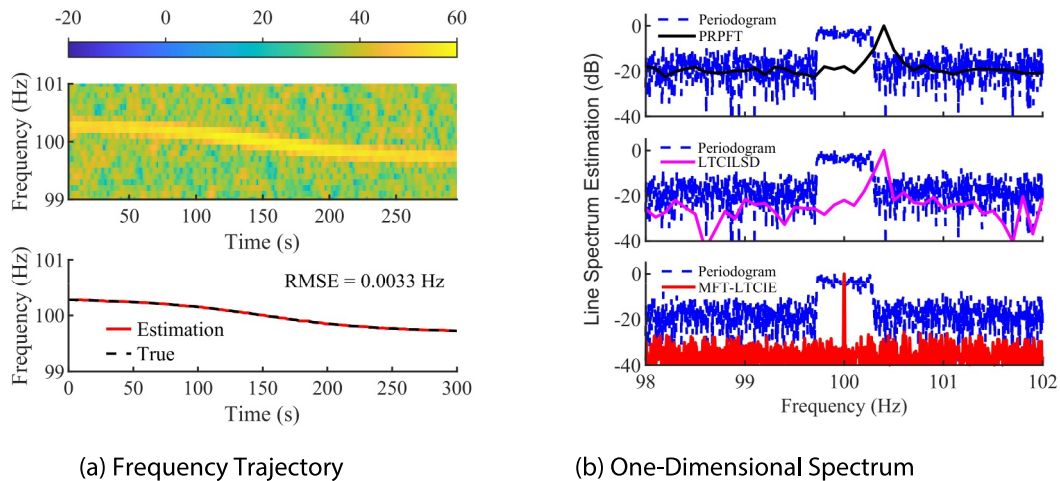


FIGURE 7 | Comparison of line-spectrum enhancement results obtained by different methods at a target velocity of 5 m/s.

obtained using different methods. Table 3 summarises the spectral peak gain and -3 dB peak width achieved by each method at different target speeds, providing a quantitative evaluation of peak prominence and spectral focussing performance.

As shown in Figure 6, when the target speed is 0 m/s, no appreciable Doppler-induced frequency drift occurs, and the line-spectrum trajectory in the LOFAR representation remains nearly horizontal. In this stationary case, the conventional periodogram already provides strong spectral concentration. Therefore, this case serves as a stationary baseline for comparison with the moving-target cases. MFT-LTCIE also maintains a

clear spectral peak, but its advantage is limited when no frequency drift is present.

When the target speed increases to 5 m/s, as shown in Figure 7, the target frequency exhibits more pronounced temporal variation, and the line-spectrum trajectory in the LOFAR representation shows a clearer Doppler-induced frequency drift. Because of the stronger Doppler variation, the conventional periodogram cannot maintain effective long-time coherent integration. As a result, the target energy spreads over a wider frequency range, appearing as a broad high-energy region rather than a narrow, well-focused spectral peak. Although PRPFT and

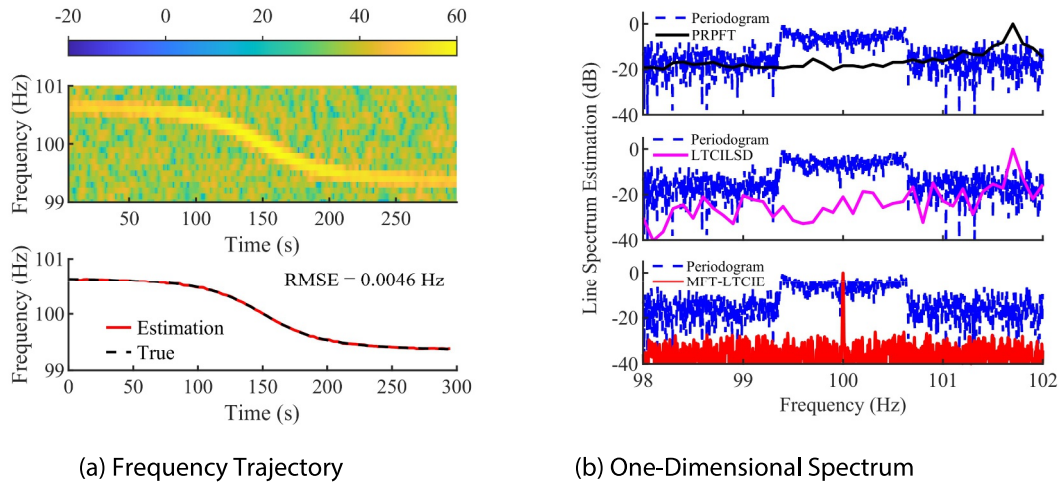


FIGURE 8 | Comparison of line-spectrum enhancement results obtained by different methods at a target velocity of 10 m/s.

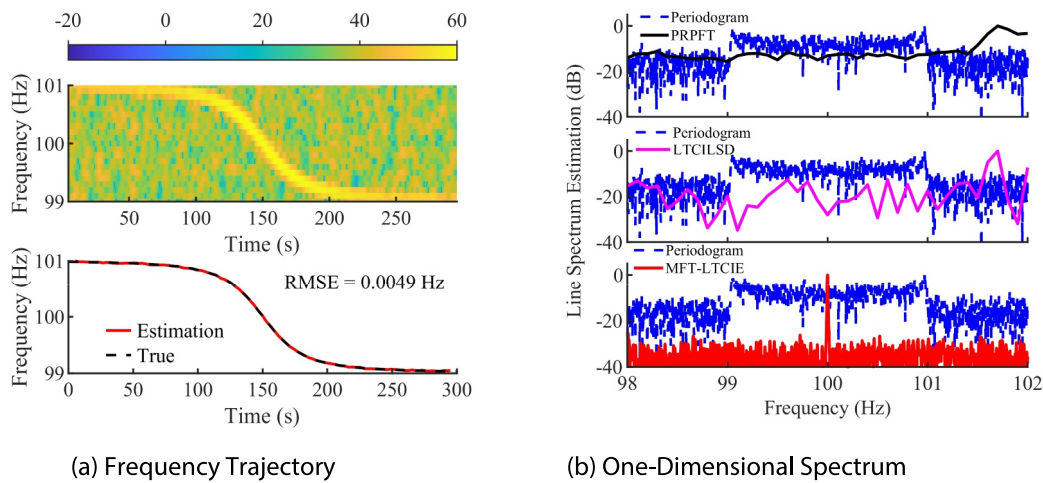


FIGURE 9 | Comparison of line-spectrum enhancement results obtained by different methods at a target velocity of 15 m/s.

TABLE 3 | Quantitative comparison of spectral peak gain and -3 dB peak width under different target velocities.

Algorithm	Peak gain/dB				-3 dB peak width/Hz			
	0 m/s	5 m/s	10 m/s	15 m/s	0 m/s	5 m/s	10 m/s	15 m/s
Periodogram	34.7	16.8	14.6	13.8	2.5×10^{-4}	> 0.5	> 1	> 1.5
PRPFT	28.8	19.5	15.6	12.7	1.0×10^{-2}	2.8×10^{-2}	3.4×10^{-2}	7.5×10^{-2}
LTCILSD	36.1	21.0	20.2	17.7	8.5×10^{-3}	1.4×10^{-2}	2.1×10^{-2}	3.2×10^{-2}
MFT-LTCIE	32.6	35.0	34.6	34.1	1.3×10^{-3}	1.5×10^{-3}	1.9×10^{-3}	2.1×10^{-3}

LTCILSD partially reduce spectral-peak broadening, noticeable local fluctuations and sidelobes remain in the enhanced spectra, limiting further spectral concentration. In contrast, MFT-LTCIE performs trajectory-guided phase compensation, refocusing the Doppler-dispersed line-spectrum energy around the target frequency. As a result, a clear and narrow main peak is obtained, indicating the improved spectral focussing capability of the proposed method.

When the target speed further increases to 10 and 15 m/s, as shown in Figures 8 and 9, Doppler variation becomes more pronounced, and the degradation of direct long-time Fourier integration becomes more severe. In these cases, the conventional periodogram mainly reflects the overall distribution of target energy and fails to form a stable, well-focused spectral peak. PRPFT and LTCILSD still provide partial line-spectrum enhancement, but their enhanced spectra contain noticeable edge peaks or local spurious peaks, and the main-peak position becomes less stable. This suggests that, under strong frequency variation, PRPFT and LTCILSD may be affected by the parameter search step size, search range and model mismatch. In contrast, MFT-LTCIE maintains a stable, well-focused spectral peak around the target frequency at both 10 and 15 m/s, indicating better robustness to strong Doppler variation.

4.3 | Quantitative Evaluation and Applicability

The peak gain is defined as the ratio between the maximum spectral magnitude at the target frequency and the average noise floor within a surrounding frequency band, expressed in decibels as follows:

$$G_{\text{peak}} = 10 \log_{10} \left(\frac{P_{\text{max}}}{P_{\text{noise}}} \right), \quad (24)$$

where P_{max} denotes the maximum spectral power, and P_{noise} represents the average power within a predefined frequency neighbourhood excluding the main lobe.

The -3 dB peak width is defined as the frequency bandwidth between the two points where the spectrum drops by 3 dB from its peak value as follows:

$$BW_{-3\text{dB}} = f_r - f_l, \quad (25)$$

where f_l and f_r are the left and right frequency points satisfying:

$$10 \log_{10}(P(f_l)) = 10 \log_{10}(P(f_r)) = 10 \log_{10}(P_{\text{max}}) - 3 \text{ dB}$$

The quantitative results in Table 3 further support the above observations. As the target speed increases from 0 to 15 m/s, the peak gain of the conventional periodogram decreases from 34.7 to 13.8 dB, whereas the -3 dB peak width increases from 2.5×10^{-4} Hz to more than 1.5 Hz. This indicates that Doppler-induced frequency variation due to target motion severely degrades the spectral focussing performance of direct long-time Fourier integration. PRPFT and LTCILSD partially mitigate spectral-peak broadening relative to the periodogram, but their peak gain generally decreases and their -3 dB peak width increases as the target speed increases, indicating degraded performance under stronger Doppler variation. In contrast, MFT-LTCIE maintains a peak gain of 32.6–35.0 dB and a -3 dB

peak width on the order of 10^{-3} Hz over the target-speed range of 0–15 m/s. In particular, at target speeds of 5–15 m/s, MFT-LTCIE achieves higher peak gain and a narrower -3 dB peak width than the other methods, indicating more stable coherent integration and better spectral focussing.

Figures 6–9 and Table 3 show that, as the target speed increases, the periodogram changes from a relatively concentrated spectral peak to pronounced peak broadening and energy dispersion, indicating that time-varying Doppler shifts degrade the coherent accumulation effect of direct long-time Fourier integration. PRPFT and LTCILSD provide partial line-spectrum enhancement at some target speeds, but their performance is affected by the parameter grid and the degree of Doppler variation. In contrast, MFT-LTCIE estimates the target frequency trajectory on a segment-wise basis and applies phase compensation guided by the estimated trajectory, refocusing the Doppler-dispersed line-spectrum energy around the target frequency. These results show that the proposed method has good robustness across different target speeds and is effective for detecting weak line spectra of moving targets over long observation intervals.

It should be noted that the performance of MFT-LTCIE depends on whether the target line-spectrum trajectory remains traceable in the LOFAR representation. When the SNR of the received signal decreases further, the target trajectory may be obscured by background-noise ridges or random local peaks, reducing the accuracy of frequency-trajectory estimation and thereby affecting phase compensation and coherent integration. Therefore, the proposed method is primarily applicable to scenarios in which the target line spectrum retains sufficient time–frequency continuity under moderate-SNR or typical low-SNR conditions. The method is mainly intended to mitigate Doppler-induced spectral broadening and energy dispersion of moving-target line spectra, rather than to address blind detection under extremely low-SNR conditions.

5 | Experimental Validation

5.1 | Experimental Description

To further assess the effectiveness of the proposed MFT-LTCIE method for enhancing line spectra of moving targets in practical scenarios, ship-radiated noise data collected during a sea trial were analysed. The sea trial was conducted in a shipping channel near Qingdao, where underwater acoustic signals radiated by passing vessels were passively recorded by a hydrophone. The vessel trajectory, speed and source–receiver range were obtained from the Automatic Identification System (AIS). The target was a noncooperative cargo ship navigating through the channel. The received acoustic signal contained narrowband line-spectrum components generated by onboard machinery and the propulsion system, providing suitable data for validating the long-time coherent integration performance under time-varying Doppler conditions.

The analyzed data segment had a duration of 600 s and covered the frequency variation near the closest point of approach (CPA) and during the subsequent receding stage. The original sampling frequency was 80 kHz. Because the target line spectra of interest were mainly distributed within approximately 988–994 Hz,

directly processing the high-rate data would introduce unnecessary computational cost. Therefore, the received signal was downsampled to 8 kHz before processing while preserving the spectral characteristics of the target line spectra. The frequency range [988, 994] Hz was adopted in the proposed MFT-LTCIE method for trajectory estimation.

Figure 10 shows the target motion geometry corresponding to the selected data segment. Figure 10a shows the target trajectory relative to the receiver, and Figure 10b shows the source–receiver range variation over time. According to the AIS data, the target travelled at an average speed of approximately 5 m/s. During the 600 s observation interval, the target first approached the receiver, reached the CPA at approximately 200 s and then gradually receded. As the radial velocity component relative to the hydrophone varied with time, the received line spectra exhibited noticeable Doppler-induced frequency shifts. In particular, as the target approached and passed the CPA, the radial velocity changed more rapidly, causing the line-spectrum trajectory to become more nonstationary.

Figure 11 shows the LOFAR representation and the uncompensated spectrum of the received signal over the selected 600 s data segment. As shown in Figure 11a, a line-spectrum trajectory is observed within approximately 988–994 Hz, and it gradually decreases over time. This behaviour indicates a pronounced Doppler-induced frequency variation caused by target motion. Figure 11b shows the uncompensated spectrum of the received

signal. Because the line-spectrum energy is distributed along a time-varying trajectory in the time–frequency plane, direct long-time Fourier integration cannot maintain phase coherence over the full observation interval. Consequently, the spectral energy is broadened and dispersed in the frequency-domain spectrum, preventing the formation of a sharp, stable spectral peak. Therefore, this data segment provides a suitable test case for evaluating frequency-trajectory estimation and line-spectrum enhancement under practical time-varying Doppler conditions.

5.2 | Experimental Results

To evaluate MFT-LTCIE under different Doppler conditions, two 200 s data segments were selected. Case 1 corresponds to an interval far from the CPA, where Doppler variation is relatively weak. This case is used to evaluate line-spectrum enhancement under weak and approximately linear Doppler variation. Case 2 covers an interval near the CPA, where Doppler variation is more pronounced, and is used to examine trajectory-compensation performance under nonlinear Doppler variation. For each case, the LOFAR representation, the frequency trajectory estimated by MFT-LTCIE, and the enhanced spectra obtained using the periodogram, PRPFT, LTCILSD and MFT-LTCIE are presented for comparison.

Figure 12 shows the results for Case 1. As shown in Figure 12a, the target is far from the CPA and recedes from the receiver during this interval, so the radial velocity relative to the receiver

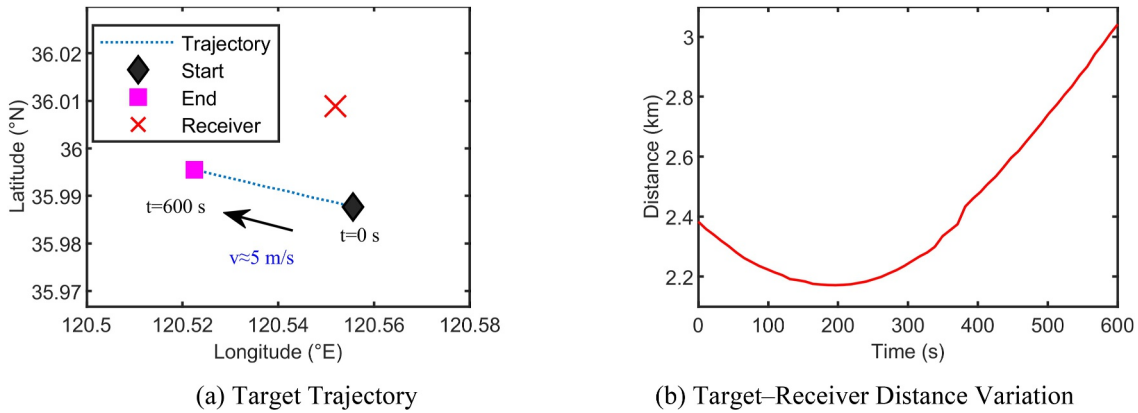


FIGURE 10 | Target motion geometry.

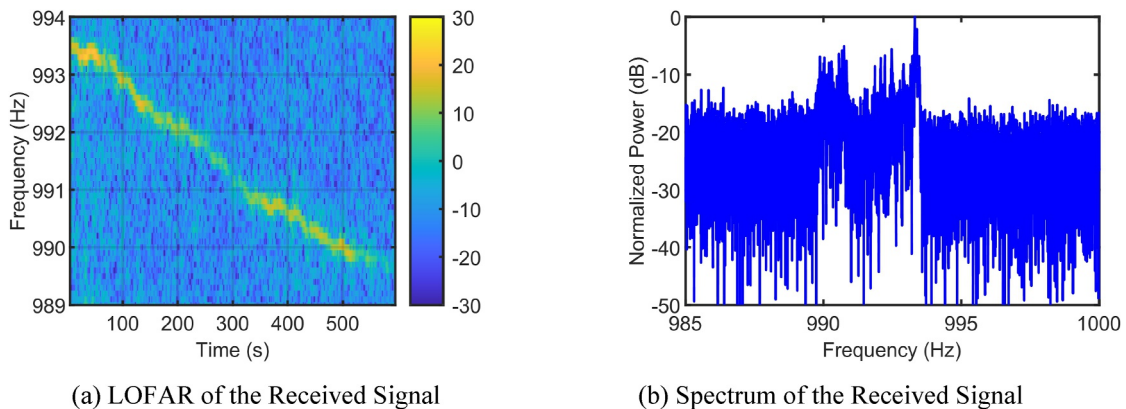


FIGURE 11 | Received signal.

changes slowly. Consequently, the Doppler-induced frequency variation is limited, and the line-spectrum trajectory follows an approximately linear trend. The trajectory estimated by MFT-LTCIE closely follows the dominant line-spectral ridge in the LOFAR representation, indicating that the proposed segment-wise matched frequency-trajectory estimation can extract local frequency variation of the target line spectrum under practical weak-Doppler conditions.

Figure 12b compares the enhanced spectra obtained using different methods. Because the conventional periodogram does not compensate for frequency drift, the target energy remains spread over a relatively wide frequency range, leading to limited spectral concentration. PRPFT and LTCILSD provide partial line-spectrum enhancement; however, because of fluctuations in the measured line spectrum and model mismatch, noticeable local fluctuations remain near the spectral peak. In contrast, MFT-LTCIE refocuses the Doppler-dispersed line-spectrum energy around the target frequency by frequency-trajectory estimation and phase compensation, producing a more focused main peak. These results indicate that, under weak and approximately linear Doppler variation far from the CPA, MFT-LTCIE improves spectral focussing and the resolvability of the target line spectrum.

Figure 13 shows the results for Case 2, which covers an interval near the CPA with nonlinear Doppler variation. As shown in Figure 13a, the target line spectrum exhibits pronounced non-stationarity in the LOFAR representation, as its frequency decreases rapidly with time. The trajectory estimated by MFT-LTCIE follows the dominant line-spectral ridge and captures the nonlinear Doppler variation. Although local fluctuations are observed, the overall estimated trajectory remains consistent with the measured line-spectral ridge, providing a reliable basis for subsequent phase compensation.

Figure 13b compares the enhanced spectra obtained using different methods. Because the conventional periodogram does not compensate for frequency drift, the spectral energy remains dispersed, and the main peak is weak. PRPFT and LTCILSD partially reduce frequency spreading; however, noticeable side-lobes and local fluctuations remain near the spectral peak. In contrast, MFT-LTCIE uses frequency-trajectory estimation and phase compensation to refocus the Doppler-dispersed line-spectrum energy, producing a more focused main peak. MFT-LTCIE therefore provides better spectral focussing than the other methods. These results indicate that, under nonlinear Doppler variation, MFT-LTCIE can restore phase coherence for long-time

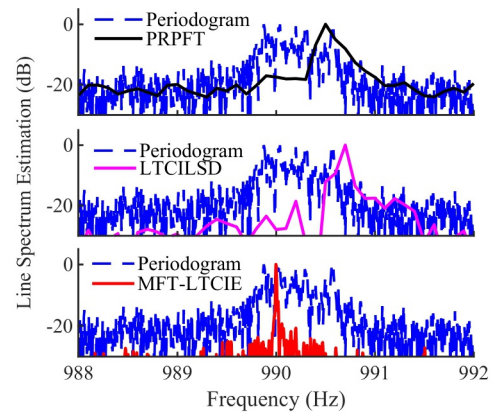
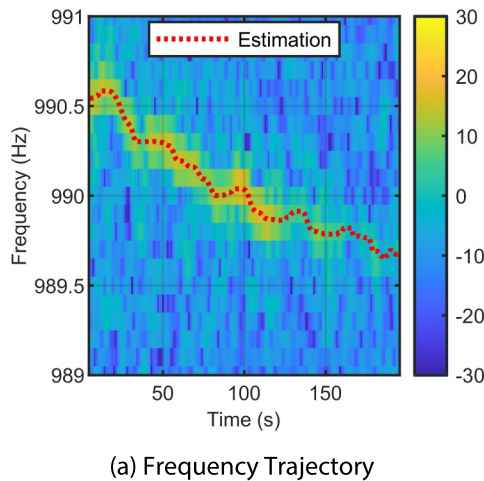


FIGURE 12 | Processing results under Case 1 conditions.

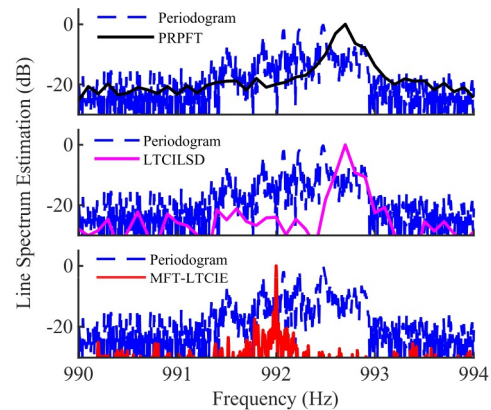
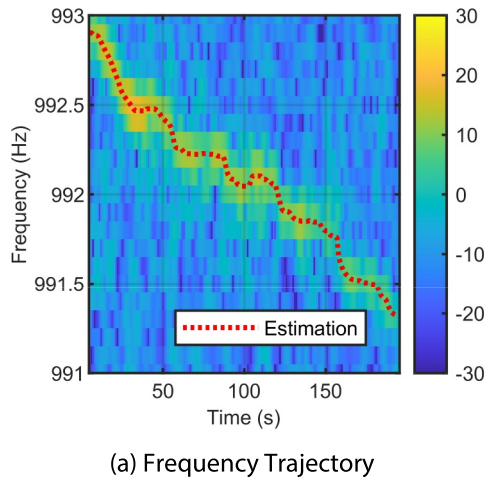


FIGURE 13 | Processing results under Case 2 conditions.

integration and improve the resolvability of the target line spectrum.

The two 200 s experimental cases show that MFT-LTCIE can enhance target line spectra under both weak, approximately linear Doppler variation and nonlinear Doppler variation near the CPA. In the segment far from the CPA, the proposed method suppresses spectral-peak broadening caused by weak frequency drift. In the segment near the CPA, it mitigates energy dispersion caused by nonlinear Doppler-induced frequency variation. These results show that the proposed method has good robustness for measured moving-target line spectra and can maintain stable frequency-trajectory estimation and line-spectrum enhancement under different Doppler-variation levels.

5.3 | Long-Time Coherent Integration Results

To further evaluate the long-time coherent integration performance of MFT-LTCIE, a 600 s data segment, denoted as Case 3, was analysed. This segment covers the continuous process in which the target approaches the receiver, passes near the CPA and then gradually recedes. The line-spectrum frequency drifts continuously over this long observation interval, placing higher demands on frequency-trajectory estimation and phase compensation.

Figure 14 shows the results for Case 3. As shown in Figure 14a, the target line-spectrum frequency gradually decreases from approximately 993 to 990 Hz, indicating pronounced long-time Doppler drift. The trajectory estimated by MFT-LTCIE closely follows the dominant line-spectral ridge. Although local fluctuations are observed because of measurement noise, propagation effects and intrinsic fluctuations of the radiated line spectrum, the overall trend remains consistent with the target frequency variation, indicating that the proposed method maintains stable frequency-trajectory estimation under long-duration observation.

Figure 14b compares the one-dimensional enhanced spectra obtained using different methods. The conventional periodogram is strongly affected by long-time frequency drift, and the target energy is dispersed over a wide frequency band, resulting

in a poorly defined spectral peak. PRPFT and LTCILSD partially improve spectral concentration, but in the presence of nonlinear frequency trajectories and local frequency jitter, spectral-peak broadening and background fluctuations remain. In contrast, MFT-LTCIE realigns the Doppler-dispersed line-spectrum energy around the reference frequency through frequency-trajectory estimation and phase compensation, producing a more focused main peak. These results indicate that MFT-LTCIE can correct phase misalignment caused by long-duration Doppler drift, thereby enhancing long-time coherent integration for measured moving-target line spectra.

6 | Conclusion

This paper has proposed a matched frequency-trajectory estimation method for long-time coherent integration enhancement (MFT-LTCIE) to mitigate spectral-peak broadening and coherent integration loss caused by time-varying Doppler shifts in moving-target line spectra over long observation intervals. Without requiring a global search over target velocity or acceleration parameters, the proposed method estimates the segment-wise frequency trajectory through short-time segmentation, candidate frequency matching and a maximum-matching criterion. By combining trajectory interpolation and phase compensation, the time-varying line-spectrum energy is aligned around a reference frequency, thereby enhancing long-time coherent integration.

Simulation results showed that MFT-LTCIE can maintain stable spectral focussing and sharp spectral peaks under different signal-to-noise ratios and target speeds. Compared with the conventional periodogram, PRPFT and LTCILSD methods, the proposed method achieved higher peak gain and a narrower -3 dB peak width. The improvement became more pronounced at higher target speeds or under stronger Doppler variation, indicating robustness to target motion and improved long-time coherent integration performance.

Experimental validation using real ship-radiated noise data further supported the effectiveness of MFT-LTCIE. The method tracked the dominant frequency variations of target line spectra under weak, approximately linear Doppler variation, nonlinear Doppler variation near the closest point of approach (CPA) and

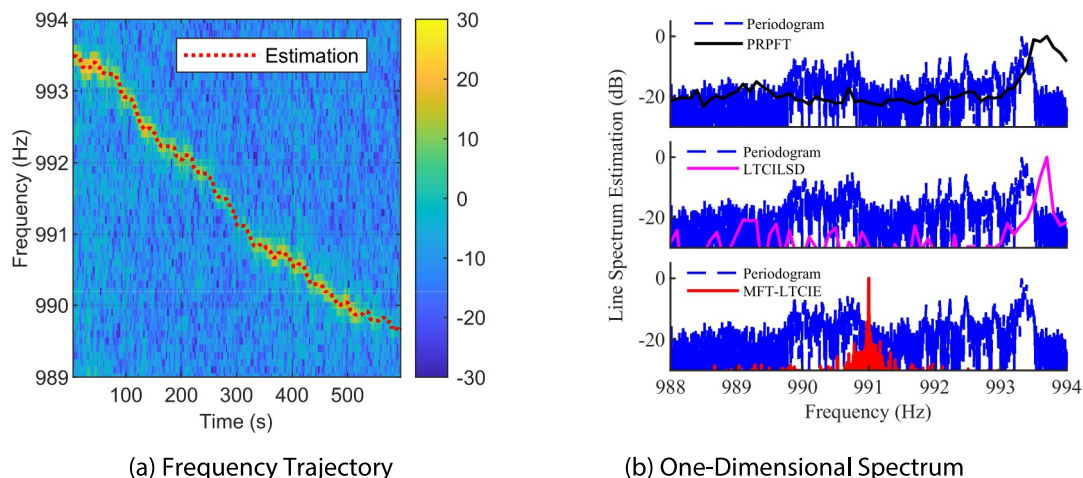


FIGURE 14 | Processing results under Case 3 conditions.

a long-duration case with an observation time of 600 s. As a result, the dispersed spectral energy was refocused around the target frequency, improving spectral-peak prominence and frequency resolution.

It should be noted that the performance of MFT-LTCIE depends on whether the target line-spectrum trajectory remains detectable in the LOFAR representation. Under extremely low-SNR or strong-interference conditions, reduced frequency-trajectory estimation accuracy may degrade phase compensation. Future work will focus on improving trajectory estimation robustness under more challenging low-SNR and strong interference conditions by exploring global optimization strategies such as dynamic programming-based methods.

Author Contributions

Yushu Zhang: writing – original draft. **Zhenxing Zhao:** conceptualization, funding acquisition, project administration. **Lin Zhai:** data curation, software. **Bo Gao:** funding acquisition, writing – review and editing. **Dazhi Gao:** funding acquisition, writing – review and editing. **Wei Gao:** writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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